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BATTERY UNIT AND WIRING UNIT FOR BATTERY

Abstract

A battery unit is provided and includes a battery and a wiring unit. The battery includes a first electrode terminal and a second electrode terminal. The wiring unit includes a flexible wiring substrate, a first coupling terminal, and a second coupling terminal. The first coupling terminal is provided on the flexible wiring substrate and is to be joined to the first electrode terminal. The second coupling terminal is provided on the flexible wiring substrate and is to be joined to the second electrode terminal. At least one of the first coupling terminal or the second coupling terminal includes an electrically conductive member and a solder layer. The electrically conductive member has a first surface, a second surface, and a through hole. The through hole extends from the first surface to the second surface. The solder layer includes a first part, a second part, and a coupling part. The first part covers all or a part of the first surface of the electrically conductive member. The second part covers all or a part of the second surface of the electrically conductive member. The coupling part passes through the through hole and couples the first part and the second part to each other.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International Application No. PCT/JP2023/045022, filed on Dec. 15, 2023, which claims priority to Japanese Patent Application No. 2023-058377, filed on Mar. 31, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure relates to a battery unit, and to a wiring unit for a battery to be used in the battery unit.

[0003] A battery unit has been known that includes a battery and a flexible wiring substrate. The battery includes a positive electrode terminal and a negative electrode terminal. The flexible wiring substrate includes a wiring coupled to the battery. A configuration of such a battery unit has been considered in various ways.

[0004] A battery unit is disclosed that includes, for example, a battery, a flexible substrate, and a fixing member. The battery includes a positive electrode terminal and a negative electrode terminal. The flexible substrate includes multiple contact terminals each of which is in contact with the corresponding one of the positive electrode terminal and the negative electrode terminal. The fixing member fixes the flexible substrate on the battery. The flexible substrate is disposed to be laid over all or a part of the battery and is fixed by the fixing member without being joined to the battery.

SUMMARY

[0005] The present disclosure relates to a battery unit, and to a wiring unit for a battery to be used in the battery unit.

[0006] Consideration has been given in various ways to improve performance of a battery unit. However, there is room for improvement in terms of the performance of the battery unit.

[0007] It is therefore desirable to provide a battery unit that allows for reduction in size and has high reliability.

[0008] The battery unit according to an embodiment of the present disclosure includes a battery and a wiring unit. The battery includes a first electrode terminal and a second electrode terminal. The wiring unit includes a flexible wiring substrate, a first coupling terminal, and a second coupling terminal. The first coupling terminal is provided on the flexible wiring substrate and is to be joined to the first electrode terminal. The second coupling terminal is provided on the flexible wiring substrate and is to be joined to the second electrode terminal. At least one of the first coupling terminal or the second coupling terminal includes an electrically conductive member and a solder layer. The electrically conductive member has a first surface, a second surface, and a through hole. The through hole extends from the first surface to the second surface. The solder layer includes a first part, a second part, and a coupling part. The first part covers all or a part of the first surface of the electrically conductive member. The second part covers all or a part of the second surface of the electrically conductive member. The coupling part passes through the through

hole and couples the first part and the second part to each other.

[0009] The battery unit of an embodiment of the present disclosure makes it possible to allow for reduction in size and to achieve high reliability.

[0010] Note that effects of the present disclosure are not necessarily limited to those described above and may include any of a series of effects described in relation to the present disclosure.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0011] FIG. 1A is a perspective diagram illustrating an overall configuration example of a battery unit according to an embodiment of the present disclosure.

[0012] FIG. 1B is a perspective diagram illustrating an external appearance of a secondary battery illustrated in FIG. 1A.

[0013] FIG. 2 is an exploded perspective diagram illustrating a configuration example of the battery unit illustrated in FIG. 1A.

[0014] FIG. 3A is a sectional diagram illustrating a sectional configuration example of the battery unit illustrated in FIG. 1A.

[0015] FIG. 3B is an enlarged sectional diagram illustrating, in an enlarged manner, a sectional configuration example of a portion of the battery unit illustrated in FIG. 3A.

[0016] FIG. 3C is an enlarged sectional diagram illustrating, in an enlarged manner, a sectional configuration example of a portion of the battery unit illustrated in FIG. 3A.

[0017] FIG. 4 is a sectional diagram illustrating a detailed configuration example of the secondary battery illustrated in FIG. 1A.

[0018] FIG. 5 is a sectional diagram illustrating a configuration example of a battery device illustrated in FIG. 4.

[0019] FIG. 6 is a perspective diagram illustrating a configuration example of an outer package can to be used in a process of manufacturing the secondary battery.

[0020] FIG. 7 is an exploded perspective diagram illustrating a configuration example of a battery unit according to an embodiment of the present disclosure.

[0021] FIG. 8A is a perspective diagram illustrating an overall configuration example of a battery unit according to an embodiment of the present disclosure.

[0022] FIG. 8B is an exploded perspective diagram illustrating an overall configuration example of the battery unit according to an embodiment of the present disclosure.

[0023] FIG. 9 is a perspective diagram illustrating a configuration example of a battery unit according to an embodiment of the present disclosure.

[0024] FIG. 10 is an exploded perspective diagram illustrating a configuration example of a wiring unit illustrated in FIG. 9 in an unbent state.

DETAILED DESCRIPTION

[0025] The present disclosure is described below in further detail including with reference to the drawings according to an embodiment.

[0026] FIG. 1A is a perspective diagram illustrating an overall configuration example of a battery unit according to an embodiment of the present disclosure. As illustrated in FIG. 1A, the battery unit includes a secondary battery 1 and a wiring unit 2 mounted on the secondary battery 1. Note that the battery unit preferably includes no circuit. As used herein, the circuit refers to a control circuit that controls, for example, an operation such as a charging operation of the secondary battery 1. Note, however, that the battery unit of the present disclosure is not limited to such a battery unit including no circuit, and may be, for example, a battery unit including a wiring unit that is provided with an electronic component including a control circuit.

[0027] The secondary battery to be described here has a flat and columnar three-dimensional shape,

and is commonly referred to as, for example, a coin type or a button type. As will be described later, the secondary battery includes two bottom parts opposed to each other, and a sidewall part positioned between the two bottom parts. The secondary battery has a height smaller than an outer diameter. The “outer diameter” is a diameter (a maximum diameter) of each of the two bottom parts. The “height” is a distance (a maximum distance) from a surface of one of the bottom parts to a surface of another of the bottom parts. Note that, in the present embodiment, a direction from one of the bottom parts toward the other of the bottom parts is assumed to be a height direction Z.

[0028] Although a charge and discharge principle of the secondary battery is not particularly limited, the following description deals with a case where a battery capacity is obtained using insertion and extraction of an electrode reactant. The secondary battery includes a positive electrode, a negative electrode, and an electrolyte. In the secondary battery, to prevent precipitation of the electrode reactant on a surface of the negative electrode during charging, a charge capacity of the negative electrode is greater than a discharge capacity of the positive electrode. In other words, an electrochemical capacity per unit area of the negative electrode is set to be greater than an electrochemical capacity per unit area of the positive electrode.

[0029] Although not particularly limited in kind, the electrode reactant is specifically a light metal such as an alkali metal or an alkaline earth metal. Examples of the alkali metal include lithium, sodium, and potassium. Examples of the alkaline earth metal include beryllium, magnesium, and calcium.

[0030] Examples are given below of a case where the electrode reactant is lithium. A secondary battery in which the battery capacity is obtained through insertion and extraction of lithium is what is called a lithium-ion secondary battery. In the lithium-ion secondary battery, lithium is inserted and extracted in an ionic state.

[0031] FIG. 1B is a perspective diagram illustrating an external appearance of the secondary battery **1** illustrated in FIG. 1A. For convenience, the following description is given with an upper side of a sheet plane of FIG. 1B assumed to be an upper side of the secondary battery, and a lower side of the sheet plane of FIG. 1B assumed to be a lower side of the secondary battery.

[0032] The secondary battery **1** to be described here has a three-dimensional shape in which a height H is smaller than an outer diameter D, as illustrated in FIG. 1B. In other words, the secondary battery **1** has a flat and columnar three-dimensional shape. Here, the three-dimensional shape of the secondary battery **1** is flat and cylindrical (circular columnar). Note that, in the present embodiment, an up-down direction on the sheet plane of FIG. 1B is assumed to be the height direction Z. Accordingly, the height H means a dimension, of the secondary battery **1**, in the height direction Z. The outer diameter D means a dimension, of the secondary battery **1**, in a direction orthogonal to the height direction Z. In other words, the outer diameter D means a dimension, of the secondary battery **1**, in a direction in a horizontal plane.

[0033] Dimensions of the secondary battery **1** are not particularly limited. However, for example, the outer diameter D is within a range from 3 mm to 30 mm both inclusive, and the height H is within a range from 0.5 mm to 70 mm both inclusive. Note that a ratio of the outer diameter D to the height H, i.e., D/H, is greater than 1. In other words, the outer diameter D is greater than the height H. Although not particularly limited, an upper limit of the ratio D/H is preferably less than or equal to 25.

[0034] The secondary battery **1** includes an upper surface **1US**, a lower surface **1LS**, and a side surface **1SS**. The lower surface **1LS** is on a side opposite to the upper surface **1US**. The side surface **1SS** couples the upper surface **1US** and the lower surface **1LS** to each other. An external terminal **20** is exposed on the upper surface **1US**. The external terminal **20** serves as a first electrode terminal (a positive electrode terminal). A bottom part **M2** of an outer package can **10** is exposed on the lower surface **1LS**. The bottom part **M2** serves as a second electrode terminal (a negative electrode terminal). The outer package can **10** includes a container part **11** and a cover part **12**. The outer package can **10** extends from the lower surface **1LS**, through the side surface **1SS**, to an outer

edge of the upper surface **1US**. The external terminal **20** is surrounded by a peripheral part **12R** (to be described later) of the cover part **12** on the upper surface **1US**. The peripheral part **12R** of the cover part **12** is a portion of the outer package can **10**.

Note that a detailed configuration of the secondary battery **1** will be described later.

[0035] FIG. **2** is an exploded perspective diagram illustrating a configuration example of the battery unit according to the present embodiment in an exploded state. The battery unit of the present embodiment further includes an insulating sheet **3** between the secondary battery **1** and the wiring unit **2**. The insulating sheet **3** is ring-shaped insulating paper having an opening **3K** in a middle region. In addition, the battery unit of the present embodiment may further include an insulating sheet **4** positioned on an opposite side of the wiring unit **2** to the secondary battery **1**, and an insulating sheet **5** positioned on an opposite side of the secondary battery **1** to the wiring unit **2**.

[0036] The wiring unit **2** includes a flexible wiring substrate **21**. The flexible wiring substrate **21** is provided with a first coupling terminal **31** and a second coupling terminal **34**. FIG. **3A** illustrates a sectional configuration example of the battery unit. FIG. **3B** is an enlarged sectional diagram illustrating, in an enlarged manner, the first coupling terminal **31** and the vicinity thereof. FIG. **3C** is an enlarged sectional diagram illustrating, in an enlarged manner, the second coupling terminal **34** and the vicinity thereof.

[0037] The first coupling terminal **31** is an electrically conductive member that is disposed to face the external terminal **20** of the secondary battery **1** and is electrically joined to the external terminal **20**. The second coupling terminal **34** is an electrically conductive member that is disposed to face the bottom part **M2** of the outer package can **10** and is electrically joined to the bottom part **M2**. The bottom part **M2** of the outer package can **10** serves as the negative electrode terminal of the secondary battery **1**.

[0038] The first coupling terminal **31** is electrically coupled to the external terminal **20** by solder. As illustrated in FIGS. **3A** and **3B**, the first coupling terminal **31** includes an electrically conductive member **32** and a solder layer **33**. The electrically conductive member **32** is a thin plate or a foil having a ring shape and including an electrically conductive material. Specifically, for example, a copper foil is usable as a material included in the electrically conductive member **32**. As illustrated in FIG. **3B**, the electrically conductive member **32** has an upper surface **32U**, a lower surface **32L**, and an end surface **32T**. The end surface **32T** couples the upper surface **32U** and the lower surface **32L** to each other. The lower surface **32L** is opposed to an upper surface of the external terminal **20**. The electrically conductive member **32** has, in a middle region thereof, a through hole **32K** extending from the upper surface **32U** to the lower surface **32L**. The end surface **32T** is an inner wall face of the through hole **32K**. The solder layer **33** is an electrically conductive layer including a first part **331**, a second part **332**, and a coupling part **333**. The first part **331** covers all or a part of the upper surface **32U** of the electrically conductive member **32**. The second part **332** covers all or a part of the lower surface **32L** of the electrically conductive member **32**. The coupling part **333** passes through the through hole **32K** and couples the first part **331** and the second part **332** to each other. The second part **332** is sandwiched between the lower surface **32L** of the electrically conductive member **32** and the upper surface of the external terminal **20** to electrically and mechanically couple the electrically conductive member **32** and the external terminal **20** to each other firmly. Here, the external terminal **20** and the first coupling terminal **31** are soldered to each other through the opening **3K** of the insulating sheet **3**. That is, at least a portion of a facing region between the external terminal **20** and the first coupling terminal **31** overlaps the opening **3K** in the height direction **Z**.

[0039] The second coupling terminal **34** is electrically coupled to the bottom part **M2** of the outer package can **10** by solder. As illustrated in FIGS. **3A** and **3C**, the second coupling terminal **34** includes an electrically conductive member **35** and a solder layer **36**. The electrically conductive member **35** is a thin plate or a foil having a ring shape and including an electrically conductive material. Specifically, for example, a copper foil is usable as a material included in the electrically

conductive member **35**. As illustrated in FIG. 3C, the electrically conductive member **35** has an upper surface **35U**, a lower surface **35L**, and an end surface **35T**. The end surface **35T** couples the upper surface **35U** and the lower surface **35L** to each other. The upper surface **35U** is opposed to a lower surface of the bottom part **M2** of the outer package can **10**. In other words, the upper surface **35U** is opposed to the lower surface **ILS** of the secondary battery **1**. The electrically conductive member **35** has, in a middle region thereof, a through hole **35K** extending from the upper surface **35U** to the lower surface **35L**. The end surface **35T** is an inner wall face of the through hole **35K**. The solder layer **36** is an electrically conductive layer including a first part **361**, a second part **362**, and a coupling part **363**. The first part **361** covers all or a part of the upper surface **35U** of the electrically conductive member **35**. The second part **362** covers all or a part of the lower surface **35L** of the electrically conductive member **35**. The coupling part **363** passes through the through hole **35K** and couples the first part **361** and the second part **362** to each other. The first part **361** is sandwiched between the lower surface of the bottom part **M2** and the upper surface **35U** of the electrically conductive member **35** to electrically and mechanically couple the bottom part **M2** and the electrically conductive member **35** to each other firmly.

[0040] As illustrated in FIG. 2, the flexible wiring substrate **21** includes a first support part **21U**, a second support part **21L**, a middle part **21M**, and an output terminal part **21C**. The first support part **21U** supports the first coupling terminal **31**. The first support part **21U** is disposed to be opposed to the upper surface **1US** of the secondary battery **1**. The second support part **21L** supports the second coupling terminal **34**. The second support part **21L** is disposed to be opposed to the lower surface **ILS** of the secondary battery **1**. The middle part **21M** couples the first support part **21U** and the second support part **21L** to each other. The middle part **21M** is provided along the side surface **1SS** of the secondary battery **1**. The output terminal part **21C** includes a first output terminal **24** and a second output terminal **27** for outputting an electromotive force of the secondary battery **1** to an outside. In the example of the wiring unit **2** illustrated in FIG. 2, the output terminal part **21C** is provided on an opposite side of the first support part **21U** to the middle part **21M**. Note that, although FIGS. 2A and 2B illustrate the flexible wiring substrate **21** in an unbent state where all the first support part **21U**, the second support part **21L**, the middle part **21M**, and the output terminal part **21C** extend along the same plane, in the battery unit of an embodiment of the present disclosure, a boundary part between the first support part **21U** and the middle part **21M** is bent substantially perpendicularly, and a boundary part between the second support part **21L** and the middle part **21M** is bent substantially perpendicularly.

[0041] As illustrated in FIG. 2, the flexible wiring substrate **21** further includes a first wiring **W1** and a second wiring **W2**. The first wiring **W1** extends from the first coupling terminal **31** to the first output terminal **24**. The second wiring **W2** extends from the second coupling terminal **34**, sequentially through the middle part **21M** and the first support part **21U**, to the second output terminal **27**. The first wiring **W1** and the second wiring **W2** are each, for example, a printed wiring including highly electrically conductive material such as copper. The first wiring **W1** and the second wiring **W2** are sandwiched between, for example, two flexible insulating films (a first layer **F1** and a second layer **F2** to be described later).

[0042] The first support part **21U** includes a first insulating member **23** that holds the first coupling terminal **31**. As illustrated in FIGS. 3A and 3B, the first insulating member **23** has a stacked structure of the first layer **F1** and the second layer **F2**. A wiring layer **22** including the first wiring **W1**, etc., is provided between the first layer **F1** and the second layer **F2**. The first layer **F1** and the second layer **F2** are each, for example, a flexible insulating film such as a polyimide film. Note that it is preferable that an outer diameter of the first support part **21U**, or in particular, an outer diameter of the first insulating member **23** be substantially equal to the outer diameter **D** of the secondary battery **1**, or be slightly smaller than the outer diameter **D**.

[0043] The middle part **21M** couples the first support part **21U** and the second support part **21L** to each other. The middle part **21M** is integrated with the first support part **21U** and the second

support part **21L**, and has a stacked structure of the first layer **F1** and the second layer **F2**. The wiring layer **22** including the second wiring **W2**, etc., is provided between the first layer **F1** and the second layer **F2**. The middle part **21M** is provided along the side surface **1SS** of a sidewall part **M3** of the secondary battery **1**.

[0044] The second support part **21L** includes a second insulating member **26** that holds the second coupling terminal **34**. As illustrated in FIGS. **3A** and **3C**, the second insulating member **26** has a stacked structure of the first layer **F1** and the second layer **F2** as with the first insulating member **23**. The wiring layer **22** including the second wiring **W2**, etc., is provided between the first layer **F1** and the second layer **F2**. Note that it is preferable that an outer diameter of the second support part **21L**, or in particular, an outer diameter of the second insulating member **26** be substantially equal to the outer diameter **D** of the secondary battery **1**, or be slightly smaller than the outer diameter **D**.

[0045] FIG. **4** is a sectional diagram illustrating a detailed configuration example of the secondary battery **1**. As illustrated in FIG. **4**, the secondary battery **1** includes the outer package can **10**, the external terminal **20**, a gasket **30**, a battery device **40**, a positive electrode lead **51**, a negative electrode lead **52**, a sealant **61**, and insulating films **62** and **63**.

[0046] The outer package can **10** is a hollow outer package member that contains the battery device **40** and other components. The outer package can **10** includes an electrically conductive material such as metal.

[0047] The outer package can **10** has a flat and circular columnar three-dimensional shape corresponding to the three-dimensional shape of the secondary battery that is flat and circular columnar. The outer package can **10** includes two bottom parts **M1** and **M2** opposed to each other, and the sidewall part **M3** positioned between the bottom parts **M1** and **M2**. In other words, the sidewall part **M3** couples the bottom part **M1** and the bottom part **M2** to each other and surrounds the battery device **40**. The sidewall part **M3** has an upper end part coupled to the bottom part **M1**. The sidewall part **M3** has a lower end part coupled to the bottom part **M2**. As described above, the outer package can **10** is substantially circular columnar. Thus, the bottom parts **M1** and **M2** are each substantially circular in plan shape, and a surface of the sidewall part **M3** is a convexly curved surface.

[0048] The outer package can **10** includes the container part **11** and the cover part **12** that are welded to each other. In other words, an internal space of the outer package can **10** is sealed by the cover part **12** being welded to the container part **11**. Note that, in the present embodiment, the bottom part **M1** forms the cover part **12**, and the bottom part **M2** and the sidewall part **M3** are integrated to form the container part **11**. Accordingly, an outer edge of the cover part **12** is welded to an end part of the sidewall part **M3** opposite to the bottom part **M2**, that is, the upper end part of the sidewall part **M3**.

[0049] The container part **11** is a container member that is to contain the battery device **40** and other components inside, and has a flat and circular columnar shape. The container part **11** has a hollow structure with an upper end part open and a lower end part closed. In other words, the container part **11** has an opening **11K** at the upper end part. The opening **11K** serves as a passing-through hole through which the battery device **40** is passable in the height direction **Z**.

[0050] As illustrated in FIG. **4**, the cover part **12** is a substantially disk-shaped cover member that closes the opening **11K** of the container part **11**, and has a through hole **12K**. The through hole **12K** is used as a coupling path for coupling the battery device **40** and the external terminal **20** to each other. The outer edge of the cover part **12** is welded to the opening **11K** of the container part **11**, as described above. The external terminal **20** is attached to the cover part **12** with the gasket **30** interposed therebetween. That is, the cover part **12** supports the external terminal **20** with the gasket **30** interposed therebetween. The external terminal **20** is attached to the cover part **12**, with the gasket **30** interposed therebetween, at a position overlapping the through hole **12K** of the cover part **12** in the height direction **Z**. The external terminal **20** is electrically insulated from the outer package can **10**.

[0051] In the secondary battery after completion, the cover part **12** is in a state of being welded to the container part **11** as described above. The opening **11K** is closed with use of the cover part **12** as described above. It may thus seem that whether the container part **11** has had the opening **11K** is no longer recognizable from an external appearance of the secondary battery.

[0052] In the secondary battery after completion, the cover part **12** is in a state of being welded

[0053] to the container part **11** as described above. The opening **11K** is closed with use of the cover part **12** as described above. It may thus seem that whether the container part **11** has had the opening **11K** is no longer recognizable from an external appearance of the secondary battery.

[0054] However, if the cover part **12** is welded to the container part **11**, welding marks remain on a surface of the outer package can **10**, more specifically, at a boundary part between the container part **11** and the cover part **12**. Thus, whether the container part **11** has had the opening **11K** is recognizable afterward based on the presence or absence of the welding marks.

[0055] Specifically, the welding marks remaining on the surface of the outer package can **10** indicates that the container part **11** has had the opening **11K**. In contrast, no welding marks remaining on the surface of the outer package can **10** indicates that the container part **11** has had no opening **11K**.

[0056] The cover part **12** is so bent as to partly protrude along the height direction Z toward an inside of the container part **11** and thus forms a recessed part **12H**. Specifically, as viewed from outside the outer package can **10**, the cover part **12** is shaped to be partly recessed in the height direction Z toward the battery device **40** contained inside the outer package can **10**. The recessed part **12H** has the through hole **12K** extending in the height direction Z, a bottom part **12HB** surrounding the through hole **12K** along a horizontal plane orthogonal to the height direction Z, and a wall part **12HW** provided upright along an outer edge of the bottom part **12HB**. A portion of the cover part **12** other than the recessed part **12H** is the peripheral part **12R**. The peripheral part **12R** is provided to surround the recessed part **12H** and has an annular shape in the horizontal plane orthogonal to the height direction Z of the secondary battery. The peripheral part **12R** is a portion that surrounds a periphery of the recessed part **12H** and protrudes away from the battery device **40** along the height direction Z. Accordingly, a surface **12HS** of the bottom part **12HB** of the recessed part **12H** is at a low position in the height direction Z toward the inside of the container part **11** as compared with a surface **12RS** of the peripheral part **12R**. In other words, a distance between the surface **12HS** of the bottom part **12HB** of the recessed part **12H** and the battery device **40** in the height direction Z is shorter than a distance between the surface **12RS** of the peripheral part **12R** and the battery device **40** in the height direction Z.

[0057] A shape of the recessed part **12H** in a plan view, that is, a shape defined by an outer edge of the recessed part **12H** when the secondary battery is viewed from above, is not particularly limited. Here, the recessed part **12H** has a substantially circular shape in a plan view. Note that an inner diameter **D12H** and a depth of the recessed part **12H** are each not particularly limited and may be set as desired. However, the depth of the recessed part **12H** is set to allow a height position of a surface **20S** of the external terminal **20** to be lower than a height position of the surface **12RS** of the peripheral part **12R**, in a state where the external terminal **20** is attached to the recessed part **12H** with the gasket **30** interposed therebetween. Additionally, the inner diameter **D12H** of the recessed part **12H** is set to be greater than an outer diameter **D20** of the external terminal **20**.

[0058] As described above, the outer package can **10** is what is called a welded can in which the container part **11** and the cover part **12** that have been physically separate from each other are welded to each other. Thus, the outer package can **10** after the welding is a single member that is physically integral as a whole, and is in a state of being not separable into the container part **11** and the cover part **12** afterward.

[0059] The outer package can **10** that is the welded can is different from a crimped can formed by crimping processing, and is what is called a crimpless can. One reason for this is to increase a device space volume inside the outer package can **10** and to thereby increase an energy density per

unit volume. The “device space volume” refers to a volume (an effective volume) of the internal space of the outer package can **10** available for containing the battery device **40**.

[0060] Further, the outer package can **10** that is the welded can does not include any portion folded over another portion, and does not include any portion in which two or more members lie over each other.

[0061] The wording “does not include any portion folded over another portion” means that the outer package can **10** is not so processed (subjected to bending processing) as to include a portion folded over another portion. The wording “does not include any portion in which two or more members lie over each other” means that the outer package can **10** after completion of the secondary battery is physically a single member and is thus not separable into two or more members afterward. That is, the outer package can **10** in the secondary battery having been completed is not in a state where two or more members lie over each other and are so combined with each other as to be separable from each other afterward.

[0062] Here, the outer package can **10** is electrically conductive. To be more specific, the container part **11** and the cover part **12** are each electrically conductive. The outer package can **10** is electrically coupled to a negative electrode **42** of the battery device **40** via the negative electrode lead **52**. Accordingly, the outer package can **10** also serves as an external coupling terminal of the negative electrode **42**. It is unnecessary for the secondary battery **1** of the present embodiment to be provided with the external coupling terminal of the negative electrode **42** separate from the outer package can **10**, which suppresses a decrease in the device space volume resulting from providing the external coupling terminal of the negative electrode **42**. As a result, the device space volume increases, and the energy density per unit volume increases accordingly.

[0063] Specifically, the outer package can **10** is a metal can that includes any one or more of electrically conductive materials including, without limitation, a metal material and an alloy material. Examples of the electrically conductive material included in the metal can include iron, copper, nickel, stainless steel, an iron alloy, a copper alloy, and a nickel alloy. The stainless steel is not particularly limited in kind, and specific examples thereof include SUS**304** and SUS**316**. Note that the container part **11** and the cover part **12** may include the same material or may include respective different materials.

[0064] The cover part **12** is insulated, via the gasket **30**, from the external terminal **20** serving as an external coupling terminal of a positive electrode **41**. One reason for this is to prevent contact, or a short circuit, between the outer package can **10** that is the external coupling terminal of the negative electrode **42** and the external terminal **20** that is the external coupling terminal of the positive electrode **41**.

[0065] As illustrated in FIGS. **1B**, **3**, and **4**, the external terminal **20** is a coupling terminal to be coupled to electronic equipment via the wiring unit **2** when the secondary battery **1** is mounted on the electronic equipment. As described above, the external terminal **20** is attached to the cover part **12** of the outer package can **10** to be supported by the cover part **12**.

[0066] The external terminal **20** is coupled to the positive electrode **41** of the battery device **40** via the positive electrode lead **51**. The external terminal **20** thus also serves as the external coupling terminal of the positive electrode **41**. Accordingly, upon use of the secondary battery **1**, the external terminal **20** serving as the external coupling terminal of the positive electrode **41** and the outer package can **10** serving as the external coupling terminal of the negative electrode **42** are coupled to electronic equipment via the wiring unit **2**. This allows the electronic equipment to operate with use of the secondary battery **1** as a power source.

[0067] The external terminal **20** is a substantially plate-shaped member that is flat and extends along a horizontal plane orthogonal to the height direction **Z** of the secondary battery. A shape of the external terminal **20** in a plan view, that is, a shape defined by an outer edge of the external terminal **20** when the secondary battery is viewed from above, is not particularly limited. In the secondary battery **1** according to the present embodiment, the shape of the external terminal **20** in a

plan view is substantially circular. The external terminal **20** is disposed inside the recessed part **12H** with the gasket **30** interposed therebetween. In other words, the external terminal **20** is provided in a state of being contained in the recessed part **12H** without protruding from the recessed part **12H** in the height direction Z. Accordingly, as illustrated in FIG. 3, the outer diameter **D20** of the external terminal **20** is smaller than the inner diameter **D12H** of the recessed part **12H** in a section along the height direction Z. Similarly, an outer diameter **D31** of the first coupling terminal **31** of the wiring unit **2** is also smaller than the inner diameter **D12H** of the recessed part **12H**. A center position of the external terminal **20** preferably match with a center line PC (to be described later) of the secondary battery.

[0068] The external terminal **20** is insulated from the cover part **12** via the gasket **30**. Here, as illustrated in FIG. 4, a position of the surface **20FS** of the external terminal **20** is low in the height direction Z toward the battery device **40** as compared with a position of the surface **12RS** of the peripheral part **12R** of the outer package can **10** in the height direction Z even at the highest position of the surface **20FS**. In the secondary battery **1** of the present embodiment, the height of the secondary battery **1** is reduced as compared with a case where the external terminal **20** protrudes above the cover part **12**. This increases the energy density per unit volume of the secondary battery **1**. This also makes it possible to prevent a short circuit between the outer package can **10** and the external terminal **20** from being caused by another electrically conductive member. The external terminal **20** includes a middle part **20C** and a peripheral part **20R**. The peripheral part **20R** surrounds the middle part. The middle part **20C** is a part of the external terminal **20** that overlaps the through hole **12K** of the cover part **12**. The positive electrode lead **51** is coupled to a back surface **20BS** of the middle part **20C**. The peripheral part **20R** overlaps the bottom part **12HB** of the recessed part **12H** in the height direction Z. Owing to the external terminal **20** and the cover part **12** having an overlap portion, it is possible to improve mechanical strength of the secondary battery as a whole.

[0069] Note that, the outer diameter **D20** of the external terminal **20** is smaller than the inner diameter **D12H** of the recessed part **12H** ($D20 < D12H$) as illustrated in FIG. 2, and thus an outer edge **20T** of the external terminal **20** is spaced from the cover part **12** as illustrated in FIG. 4. Accordingly, the gasket **30** is disposed in only a portion of a region between the external terminal **20** and the cover part **12** (the recessed part **12H**). More specifically, the gasket **30** is disposed only at a location where the external terminal **20** and the cover part **12** would be in contact with each other if it were not for the gasket **30**. However, the gasket **30** may also be provided between an inner wall face of the wall part **12HW** of the recessed part **12H** and the outer edge **20T** of the external terminal **20**.

[0070] The external terminal **20** includes any one or more of electrically conductive materials including, without limitation, a metal material and an alloy material. The external terminal **20** may include a single layer or may include a stacked body including two or more layers having different linear expansion coefficients from each other, for example. Specifically, the external terminal **20** may be a stacked body of a first layer including Ni (nickel), a second layer including stainless steel such as SUS304, and a third layer including Al (aluminum).

[0071] The gasket **30** is an insulating member disposed between the outer package can **10** (the cover part **12**) and the external terminal **20** as illustrated in FIG. 2. The external terminal **20** is fixed to the cover part **12** with the gasket **30** interposed therebetween. The gasket **30** is ring-shaped in a plan view and has a through hole at a location corresponding to the through hole **12K**. The gasket **30** includes any one or more of insulating materials including, without limitation, a polymer compound having an insulating property. The insulating materials are resins including, without limitation, polypropylene and polyethylene.

[0072] A range of placement of the gasket **30** may be chosen as desired. Here, the gasket **30** is disposed in a gap between the surface **12HS** of the bottom part **12HB** of the recessed part **12H** and the back surface **20BS** of the external terminal **20**. However, as described above, the gasket **30** is

preferably also provided between the inner wall face of the wall part **12HW** of the recessed part **12H** and the outer edge **20T** of the external terminal **20**. Further, the cover part **12** and the external terminal **20** are preferably stuck to each other by the gasket **30**.

[0073] The battery device **40** is a power generation device that causes charging and discharging reactions to proceed. As illustrated in FIGS. **3** and **4**, the battery device **40** is contained inside the outer package can **10**. As illustrated in FIG. **4**, the battery device **40** includes the positive electrode **41** and the negative electrode **42**. The positive electrode **41** serves as a first electrode. The negative electrode **42** serves as a second electrode. Here, the battery device **40** further includes a separator **43** and an electrolytic solution. The electrolytic solution is a liquid electrolyte.

[0074] The center line PC illustrated in FIG. **4** is a line segment corresponding to a center of the battery device **40** in a direction along the outer diameter D of the secondary battery (the outer package can **10** of the secondary battery). More specifically, a position P of the center line PC corresponds to a position of the center of the battery device **40**.

[0075] The battery device **40** is what is called an electrode wound body. More specifically, in the battery device **40**, the positive electrode **41** and the negative electrode **42** are stacked on each other with the separator **43** interposed between the positive electrode **41** and the negative electrode **42**. In addition, the stack of the positive electrode **41**, the negative electrode **42**, and the separator **43** is wound around the center line PC that is a winding axis. The positive electrode **41** and the negative electrode **42** are wound, remaining in a state of being opposed to each other with the separator **43** interposed between the positive electrode **41** and the negative electrode **42**. As a result, a winding center space **40K** serving as an internal space is present at the center of the battery device **40**.

[0076] Here, the positive electrode **41**, the negative electrode **42**, and the separator **43** are so wound that the separator **43** is disposed in each of an outermost wind of the electrode wound body and an innermost wind of the electrode wound body. Respective numbers of winds of the positive electrode **41**, the negative electrode **42**, and the separator **43** are not particularly limited, and may be chosen as desired.

[0077] The battery device **40** has a three-dimensional shape based on the three-dimensional shape of the outer package can **10**. Specifically, the battery device **40** has a flat and circular columnar three-dimensional shape. This helps to prevent what is called a dead space, more specifically, a gap between the outer package can **10** and the battery device **40**, from easily being provided when the battery device **40** is placed inside the outer package can **10**, as compared with a case where the battery device **40** has a three-dimensional shape different from the three-dimensional shape of the outer package can **10**. This allows for efficient use of the internal space of the outer package can **10**. As a result, the device space volume increases, and the energy density per unit volume of the secondary battery increases accordingly.

[0078] FIG. **5** is a sectional diagram illustrating a configuration example of a portion of the battery device **40**. The positive electrode **41** is the first electrode to be used to cause the charging and discharging reactions to proceed. As illustrated in FIG. **5**, the positive electrode **41** includes a positive electrode current collector **41A** and a positive electrode active material layer **41B**.

[0079] The positive electrode current collector **41A** has two opposed surfaces on each of which the positive electrode active material layer **41B** is to be provided. The positive electrode current collector **41A** includes an electrically conductive material such as a metal material. Examples of the metal material include aluminum.

[0080] The positive electrode active material layer **41B** is provided on each of the two opposed surfaces of the positive electrode current collector **41A**. The positive electrode active material layer **41B** includes any one or more of positive electrode active materials into which lithium is insertable and from which lithium is extractable. Note that the positive electrode active material layer **41B** may be provided only on one of the two opposed surfaces of the positive electrode current collector **41A**. The positive electrode active material layer **41B** may further include other materials including, without limitation, a positive electrode binder and a positive electrode conductor. A

method of forming the positive electrode active material layer **41B** is not particularly limited, and specific examples thereof include a coating method.

[0081] The positive electrode active material includes a lithium compound. The term “lithium compound” is a generic term for a compound that includes lithium as a constituent element. More specifically, the lithium compound is a compound that includes lithium and one or more transition metal elements as constituent elements. One reason for this is that a high energy density is obtainable. Note that the lithium compound may further include any one or more of other elements (excluding lithium and transition metal elements). Although not particularly limited in kind, the lithium compound is specifically an oxide, a phosphoric acid compound, a silicic acid compound, or a boric acid compound, for example. Specific examples of the oxide include $\text{LiNiO}_{0.2}$, $\text{LiCoO}_{0.2}$, and $\text{LiMn}_{0.2}\text{O}_{0.4}$. Specific examples of the phosphoric acid compound include $\text{LiFePO}_{0.4}$ and $\text{LiMnPO}_{0.4}$.

[0082] The positive electrode binder includes any one or more of materials including, without limitation, a synthetic rubber and a polymer compound. Examples of the synthetic rubber include a styrene-butadiene-based rubber. Examples of the polymer compound include polyvinylidene difluoride. The positive electrode conductor includes any one or more of electrically conductive materials including, without limitation, a carbon material. Examples of the carbon material include graphite, carbon black, acetylene black, and Ketjen black. Note that the electrically conductive material may be a metal material or a polymer compound, for example.

[0083] The negative electrode **42** is the second electrode to be used to cause the charging and discharging reactions to proceed. As illustrated in FIG. 3, the negative electrode **42** includes a negative electrode current collector **42A** and a negative electrode active material layer **42B**.

[0084] The negative electrode current collector **42A** has two opposed surfaces on each of which the negative electrode active material layer **42B** is to be provided. The negative electrode current collector **42A** includes an electrically conductive material such as a metal material. Examples of the metal material include copper.

[0085] The negative electrode active material layer **42B** is provided on each of the two opposed surfaces of the negative electrode current collector **42A**. The negative electrode active material layer **42B** includes any one or more of negative electrode active materials into which lithium is insertable and from which lithium is extractable. Note that the negative electrode active material layer **42B** may be provided only on one of the two opposed surfaces of the negative electrode current collector **42A**. The negative electrode active material layer **42B** may further include other materials including, without limitation, a negative electrode binder and a negative electrode conductor. Details of the negative electrode binder are similar to the details of the positive electrode binder. Details of the negative electrode conductor are similar to the details of the positive electrode conductor. A method of forming the negative electrode active material layer **42B** is not particularly limited, and specifically includes any one or more of methods including, without limitation, a coating method, a vapor-phase method, a liquid-phase method, a thermal spraying method, and a firing (sintering) method.

[0086] The negative electrode active material includes a carbon material, a metal-based material, or both. One reason for this is that a high energy density is obtainable. Examples of the carbon material include graphitizable carbon, non-graphitizable carbon, and graphite (natural graphite and artificial graphite). The metal-based material is a material that includes, as one or more constituent elements, any one or more elements among metal elements and metalloid elements that are each able to form an alloy with lithium. Examples of such metal elements and metalloid elements include silicon, tin, or both. The metal-based material may be a simple substance, an alloy, a compound, a mixture of two or more thereof, or a material including two or more phases thereof. Specific examples of the metal-based material include $\text{TiSi}_{0.2}$ and $\text{SiO}_{0.2}$ ($0 < x \leq 2$ or $0.2 < x < 1.4$).

[0087] Here, the negative electrode **42** has a height greater than a height of the positive electrode

41. More specifically, the negative electrode **42** protrudes above the positive electrode **41**, and protrudes below the positive electrode **41**. One reason for this is to prevent precipitation of lithium extracted from the positive electrode **41**. The “height” is a dimension corresponding to the height **H** of the secondary battery described above, that is, a dimension in the up-down direction in each of FIGS. **1B** and **2**. The definition of the height described here applies also to the following.

[0088] The separator **43** is an insulating porous film interposed between the positive electrode **41** and the negative electrode **42**, as illustrated in FIGS. **4** and **5**. The separator **43** allows lithium ions to pass through the separator **43** and prevents a short circuit between the positive electrode **41** and the negative electrode **42**. The separator **43** includes a polymer compound such as polyethylene.

[0089] Here, the separator **43** has a height greater than the height of the negative electrode **42**. More specifically, the separator **43** preferably protrudes above the negative electrode **42** and protrudes below the negative electrode **42**. One reason for this is to insulate the positive electrode lead **51** from the negative electrode **42** by using the separator **43**.

[0090] The electrolytic solution includes a solvent and an electrolyte salt. The positive electrode **41**, the negative electrode **42**, and the separator **43** are each impregnated with the electrolytic solution. The solvent includes any one or more of non-aqueous solvents (organic solvents) including, without limitation, a carbonic-acid-ester-based compound, a carboxylic-acid-ester-based compound, and a lactone-based compound. An electrolytic solution that includes any of the non-aqueous solvents is what is called a non-aqueous electrolytic solution. The electrolyte salt includes any one or more of light metal salts including, without limitation, a lithium salt.

[0091] As illustrated in FIG. **4**, the positive electrode lead **51** is contained inside the outer package can **10**. The positive electrode lead **51** is a coupling wiring coupled to each of the positive electrode **41** and the external terminal **20**. The secondary battery illustrated in FIG. **2** includes one positive electrode lead **51**. However, the secondary battery may include two or more positive electrode leads **51**.

[0092] The positive electrode lead **51** is coupled to an upper end part of the positive electrode **41**. Specifically, the positive electrode lead **51** is coupled to an upper end part of the positive electrode current collector **41A**. Further, the positive electrode lead **51** is coupled to a lower surface **20LS** of the external terminal **20** through the through hole **12K** provided in the cover part **12**. A method of coupling the positive electrode lead **51** is not particularly limited, and specifically includes any one or more of welding methods including, without limitation, a resistance welding method and a laser welding method. The details of the welding methods described here apply also to the following.

[0093] A portion of the positive electrode lead **51** is electrically insulated from each of the cover part **12** of the outer package can **10** and the negative electrode **42** of the battery device **40**, and is sandwiched by the cover part **12** and the battery device **40** in the height direction of the secondary battery. As illustrated in FIG. **2**, the positive electrode lead **51** includes a first part **511**, a second part **512**, and a turning part **513**. The first part **511** and the second part **512** each extend along a horizontal plane orthogonal to the height direction **Z** of the secondary battery. Further, the first part **511** and the second part **512** overlap each other in the height direction **Z** of the secondary battery, with the sealant **61** interposed between the first part **511** and the second part **512**. The turning part **513** is so curved as to couple the first part **511** and the second part **512** to each other.

[0094] The first part **511** and the second part **512** are sandwiched between the battery device **40** and a protruding part **12P** of the cover part **12** in the height direction **Z** of the secondary battery.

[0095] In this way, the portion of the positive electrode lead **51** is held by the cover part **12** and the battery device **40** by extending along each of a lower surface of the cover part **12** and an upper surface of the battery device **40**. This allows the positive electrode lead **51** to be fixed inside the outer package can **10**. By preventing the positive electrode lead **51** from easily moving even if the secondary battery experiences an external force such as vibration or impact, the positive electrode lead **51** is prevented from being easily damaged. Examples of damage to the positive electrode lead **51** include cracking of the positive electrode lead **51**, breakage of the positive electrode lead **51**,

and detachment of the positive electrode lead **51** from the positive electrode **41**.

[0096] More specifically, the wording “a portion of the positive electrode lead **51** is sandwiched by the outer package can **10** and the battery device **40**” means that the positive electrode lead **51** is held by the outer package can **10** and the battery device **40** from above and below while being insulated from each of the outer package can **10** and the battery device **40**, and that the positive electrode lead **51** is thus in a state of being not easily movable inside the outer package can **10** even if the secondary battery experiences an external force such as vibration or impact. The state where the positive electrode lead **51** is not easily movable inside the outer package can **10** exactly indicates that the battery device **40** is also in the state of being not easily movable inside the outer package can **10**. This helps to also suppress a defect of the battery device **40**, i.e., the electrode wound body, such as winding deformation when the secondary battery experiences vibration or impact.

[0097] Note that the positive electrode lead **51** may be in a state of being partially embedded in the battery device **40** because of being pressed by the battery device **40**. More specifically, the positive electrode lead **51** may be in a state of being partially embedded in an upper end part of the separator **43** because of the height of the separator **43** being greater than the height of each of the positive electrode **41** and the negative electrode **42** as described above. In such a case, a recessed part is formed in the upper end part of the separator **43** because of being pressed by the positive electrode lead **51**. All or a part of the positive electrode lead **51** is received in the recessed part, which allows the positive electrode lead **51** to be held by the separator **43**. This further prevents the positive electrode lead **51** from easily moving inside the outer package can **10**, and to thereby further prevent the positive electrode lead **51** from being easily damaged.

[0098] Here, as described above, the cover part **12** includes the protruding part **12P**, and a portion of the positive electrode lead **51** is sandwiched by the protruding part **12P** and the battery device **40**. More specifically, a portion of the positive electrode lead **51** is held by the protruding part **12P** and the battery device **40** by extending along each of a lower surface of the protruding part **12P** and the upper surface of the battery device **40**. The protruding part **12P** helps to hold the positive electrode lead **51** more easily. This further prevents the positive electrode lead **51** from being easily damaged.

[0099] Further, a portion of the positive electrode lead **51** is insulated from the cover part **12** and the negative electrode **42** via each of the separator **43**, the sealant **61**, and the insulating film **62**.

[0100] Specifically, as described above, the height of the separator **43** is greater than the height of the negative electrode **42**. Accordingly, a portion of the positive electrode lead **51** is separate from the negative electrode **42** via the separator **43**, and is thus insulated from the negative electrode **42** via the separator **43**. One reason for this is to prevent a short circuit between the positive electrode lead **51** and the negative electrode **42**.

[0101] Further, the positive electrode lead **51** is covered at a periphery thereof by the sealant **61** having an insulating property. A portion of the positive electrode lead **51** is thus insulated from each of the cover part **12** and the negative electrode **42** via the sealant **61**. One reason for this is to prevent a short circuit between the positive electrode lead **51** and the cover part **12**, and to also prevent a short circuit between the positive electrode lead **51** and the negative electrode **42**.

[0102] Further, the insulating film **62** is disposed between the cover part **12** and the positive electrode lead **51**. A portion of the positive electrode lead **51** is thus insulated from the cover part **12** via the insulating film **62**. One reason for this is to prevent a short circuit between the positive electrode lead **51** and the cover part **12**.

[0103] Furthermore, the insulating film may also be disposed between the battery device **40** and the positive electrode lead **51**.

[0104] Details of a material included in the positive electrode lead **51** are similar to the details of the material included in the positive electrode current collector **41A**. Note that the material included in the positive electrode lead **51** and the material included in the positive electrode current

collector **41A** may be the same as or different from each other.

[0105] Here, the positive electrode lead **51** is coupled to the positive electrode **41** in a region on a front side relative to the center line PC, i.e., a region on a right side relative to the center line PC in FIG. **4**. In order to be coupled to the external terminal **20**, the positive electrode lead **51** includes the turning part **513** in the middle of extension to the external terminal **20**. The turning part **513** is present in a region on a back side relative to the center line PC, i.e., a region on a left side relative to the center line PC in FIG. **4**. The positive electrode lead **51** includes the first part **511** that corresponds to a part extending from a location where the positive electrode lead **51** is coupled to the positive electrode **41**, through the center position P, to the turning part **513**. The first part **511** extends along the upper surface of the battery device **40** in a direction orthogonal to the height direction Z. In addition, the positive electrode lead **51** includes the second part **512** that corresponds to a part in the middle of extension from the turning part **513** to a location where the positive electrode lead **51** is coupled to the external terminal **20**. The second part **512** extends along the upper surface of the battery device **40** in the direction orthogonal to the height direction Z to be laid over the first part **511**. As described above, a portion of the positive electrode lead **51** is sandwiched by the cover part **12** and the battery device **40** and extends toward the external terminal **20**, in both the region on the front side relative to the center line PC and the region on the back side relative to the center line PC.

[0106] Here, as is apparent from FIG. **4**, “the region on the front side relative to the center line PC” is, where the battery device **40** is divided into two regions with respect to the center line PC in a direction along the outer diameter D, one region in which the location where the positive electrode lead **51** is coupled to the positive electrode **41** is present. In FIG. **4**, “the region on the front side relative to the center line PC” is the region on the right side relative to the center line PC. In contrast, as is apparent from FIG. **4**, “the region on the back side relative to the center line PC” is another region of the two regions, and is the region on the left side relative to the center line PC in FIG. **4**. In other words, “the region on the back side relative to the center line PC” is, where the battery device **40** is divided into the two regions with respect to the center line PC in the direction along the outer diameter D, the other region in which the location where the positive electrode lead **51** is coupled to the positive electrode **41** is absent.

[0107] A position of coupling of the positive electrode lead **51** to the positive electrode **41** is not particularly limited, and may be chosen as desired. In particular, the positive electrode lead **51** is preferably coupled to the positive electrode **41** on an inner side of winding of the positive electrode **41** relative to an outermost wind of the positive electrode **41**. One reason for this is that corrosion of the outer package can **10** caused by creeping up of the electrolytic solution is suppressed unlike when the positive electrode lead **51** is coupled to the positive electrode **41** in the outermost wind of the positive electrode **41**. The “creeping up of the electrolytic solution” refers to a phenomenon in which, when the positive electrode lead **51** is disposed adjacent to an inner wall face of the outer package can **10**, the electrolytic solution in the battery device **40** creeps up along the positive electrode lead **51** to reach the inner wall face of the outer package can **10**. The electrolytic solution coming into contact with the outer package can **10** as a result of the “creeping up of the electrolytic solution” causes a phenomenon in which the outer package can **10** dissolves or changes in color.

[0108] Here, in a region between the positive electrode **41** and the external terminal **20**, the positive electrode lead **51** is turned up once or more and thus lies over itself once or more. The number of times the positive electrode lead **51** is to be turned up is not particularly limited as long as it is once or more. The wording “the positive electrode lead **51** is turned up” means that the extending direction of the positive electrode lead **51** changes at an angle greater than 90° in the middle of the positive electrode lead **51**. The positive electrode lead **51** preferably has, at a location where the positive electrode lead **51** is turned up, a curved shape rather than a bent shape, as with the turning part **513**. Further, although FIG. **4** illustrates an example in which the positive electrode lead **51** includes one turning part **513**, the positive electrode lead **51** may include multiple turning parts

513.

[0109] The positive electrode lead **51** is turned up at the turning part **513** in the middle of extension from the positive electrode **41** to the external terminal **20**. Specifically, as illustrated in FIG. 2, the first part **511** extends from a first position **P1** to a second position **P2** in a horizontal plane orthogonal to the height direction of the secondary battery. The first position **P1** is other than the center position **P** of the outer package can **10**. The second position **P2** is on an opposite side of the center position to the first position **P1**. The second part **512** extends from the second position **P2** toward the center position **P**. In the positive electrode lead **51**, an overlap portion of the first part **511** and the second part **512** is a surplus portion. It can thus be said that the positive electrode lead **51** has a length margin in a longitudinal direction of the positive electrode lead **51**.

[0110] This provides room to change an orientation of the cover part **12** relative to the container part **11** when forming the outer package can **10** by using the container part **11** and the cover part **12** in a process of manufacturing the secondary battery, as will be described later. Further, when the secondary battery experiences an external force such as vibration or impact, the length margin of the positive electrode lead **51** is usable to mitigate the external force, thereby helping to prevent the positive electrode lead **51** from being easily damaged. Furthermore, the length margin of the positive electrode lead **51** is usable to change the position of coupling of the positive electrode lead **51** to the positive electrode **41** to a desired position without changing the positive electrode lead **51** in length.

[0111] In this case, the length (an entire length including the length margin) of the positive electrode lead **51** is not particularly limited, and may be chosen as desired. The length of the positive electrode lead **51** is preferably greater than or equal to half the outer diameter **D** of the outer package can **10**, in particular. One reason for this is to ensure that the length of the positive electrode lead **51** has a length margin allowing for raising the cover part **12** relative to the container part **11**, and to thereby make it easier to raise the cover part **12** relative to the container part **11**.

[0112] A range of coupling of the positive electrode lead **51** to the external terminal **20** is not particularly limited. It is preferable that the range of coupling of the positive electrode lead **51** to the external terminal **20** be wide enough for the positive electrode lead **51** to be prevented from easily becoming detached from the external terminal **20** and be narrow enough to allow for the length margin of the positive electrode lead **51**, in particular. One reason why the range of coupling of the positive electrode lead **51** to the external terminal **20** is preferably narrow enough is that a sufficiently large length margin of the positive electrode lead **51** is achievable because a portion of the positive electrode lead **51** not coupled to the external terminal **20** serves as the length margin.

[0113] Note that the positive electrode lead **51** is provided separately from the positive electrode current collector **41A**. However, the positive electrode lead **51** may be physically continuous with the positive electrode current collector **41A** and may thus be integrated with the positive electrode current collector **41A**.

[0114] As illustrated in FIG. 4, the negative electrode lead **52** is contained inside the outer package can **10**. The negative electrode lead **52** is coupled to each of the negative electrode **42** and the outer package can **10** (the container part **11**). Here, the secondary battery includes one negative electrode lead **52**. However, the secondary battery may include two or more negative electrode leads **52**.

[0115] The negative electrode lead **52** is coupled to a lower end part of the negative electrode **42**, more specifically, a lower end part of the negative electrode current collector **42A**. Further, the negative electrode lead **52** is coupled to a bottom surface of the container part **11**. Details of a method of coupling the negative electrode lead **52** are similar to the details of the method of coupling the positive electrode lead **51**.

[0116] Details of a material included in the negative electrode lead **52** are similar to the details of the material included in the negative electrode current collector **42A**. Note that the material included in the negative electrode lead **52** and the material included in the negative electrode current collector **42A** may be the same as or different from each other.

[0117] A position of coupling of the negative electrode lead **52** to the negative electrode **42** is not particularly limited, and may be chosen as desired. Here, the negative electrode lead **52** is coupled to an outermost wind part of the negative electrode **42** included in the electrode wound body.

[0118] Note that the negative electrode lead **52** is provided separately from the negative electrode current collector **42A**. However, the negative electrode lead **52** may be physically continuous with the negative electrode current collector **42A** and may thus be integrated with the negative electrode current collector **42A**.

[0119] The sealant **61** is a first insulating member covering the periphery of the positive electrode lead **51**, as illustrated in FIG. **4**. The sealant **61** includes two insulating tapes each being attached to corresponding one of a front surface and a back surface of the positive electrode lead **51**. Here, to allow the positive electrode lead **51** to be coupled to each of the positive electrode **41** and the external terminal **20**, the sealant **61** covers the periphery of a portion in the middle of the positive electrode lead **51**. Note that a structure of the sealant **61** is not limited to a tape-shaped structure, and the sealant **61** may have a tube-shaped structure, for example.

[0120] The sealant **61** includes any one or more of insulating materials including, without limitation, a polymer compound having an insulating property. Examples of the insulating materials include polyimide.

[0121] The insulating film **62** is disposed between the cover part **12** and the positive electrode lead **51** in the height direction **Z**, as illustrated in FIG. **4**. Here, the insulating film **62** is ring-shaped in a plan view and has an opening **62K** at a location corresponding to the through hole **12K** in the height direction **Z**.

[0122] The insulating film **62** may have an unillustrated adhesive layer on one surface, and may thus be adhered to either the cover part **12** or the positive electrode lead **51** via the adhesive layer. Alternatively, the insulating film **62** may have respective adhesive layers on both surfaces, and may thus be adhered to both the cover part **12** and the positive electrode lead **51** via the respective adhesive layers.

[0123] The insulating film **62** may include any one or more of insulating materials including, without limitation, a polymer compound having an insulating property. Examples of the one or more insulating materials to be included in the insulating film **62** include polyimide.

[0124] The insulating film **63** is a third insulating member disposed between the battery device **40** and the positive electrode lead **51**, as illustrated in FIG. **4**. Here, the insulating film **63** is flat plate-shaped in a plan view. The insulating film **63** is disposed to close the winding center space **40K** and to cover the battery device **40** around the winding center space **40K**.

[0125] Details of a material included in the insulating film **63** are similar to the details of the material included in the insulating film **62**. Note that the material included in the insulating film **63** and the material included in the insulating film **62** may be the same as or different from each other.

[0126] Note that the secondary battery **1** may further include one or more other components according to an embodiment.

[0127] Specifically, the secondary battery **1** includes a safety valve mechanism. The safety valve mechanism is to cut off electrical coupling between the outer package can **10** and the battery device **40** if an internal pressure of the outer package can **10** reaches a certain level or higher. Examples of a factor that causes the internal pressure of the outer package can **10** to reach the certain level or higher include the occurrence of a short circuit inside the secondary battery **1** and heating of the secondary battery from outside. Although a placement location of the safety valve mechanism is not particularly limited, the safety valve mechanism is preferably placed on either the bottom part **M1** or the bottom part **M2**, and more preferably, on the bottom part **M2** to which no external terminal **20** is attached, in particular.

[0128] Further, the secondary battery **1** may include an insulator between the outer package can **10** and the battery device **40**. The insulator includes any one or more of materials including, without limitation, an insulating film and an insulating sheet, and prevents a short circuit between the outer

package can **10** and the battery device **40**. A range of placement of the insulator is not particularly limited, and may be chosen as desired.

[0129] Note that the outer package can **10** is provided with a cleavage valve. The cleavage valve cleaves to release the internal pressure of the outer package can **10** when the internal pressure reaches a certain level or higher. A placement location of the cleavage valve is not particularly limited. However, the cleavage valve is preferably placed on either the bottom part **M1** or the bottom part **M2**, and more preferably, on the bottom part **M2**, in particular, as with the placement location of the safety valve mechanism described above.

[0130] Upon charging of the secondary battery **1**, in the battery device **40**, lithium is extracted from the positive electrode **41**, and the extracted lithium is inserted into the negative electrode **42** through the electrolytic solution. Upon discharging of the secondary battery, in the battery device **40**, lithium is extracted from the negative electrode **42**, and the extracted lithium is inserted into the positive electrode **41** through the electrolytic solution. Upon the charging and the discharging, lithium is inserted and extracted in an ionic state.

[0131] A description is given next of a method of manufacturing the battery unit according to an embodiment.

[0132] A description is given first of a method of manufacturing the secondary battery **1**. FIG. **6** illustrates a perspective configuration of the outer package can **10** to be used in the process of manufacturing the secondary battery **1**, and corresponds to FIG. **1B**.

[0133] FIG. **6** illustrates a state where the cover part **12** is separate from the container part **11** before the cover part **12** is welded to the container part **11**. In the following description, where appropriate, FIGS. **1A** to **5** described already will be referred to in conjunction with FIG. **6**.

[0134] Here, as illustrated in FIG. **6**, the container part **11** and the cover part **12** that are physically separate from each other are prepared to form the outer package can **10**. The container part **11** is a substantially bowl-shaped member in which a bottom part **11B** and a sidewall part **11W** are integrated with each other, and has the opening **11K**. Alternatively, the bottom part **11B** and the sidewall part **11W** that are physically separate from each other may be prepared, and the sidewall part **11W** may be welded to the bottom part **11B** to thereby form the container part **11**. A convex part **11P** is provided on an inner side of the sidewall part **11W**. The convex part **11P** is formed integrally with, for example, the sidewall part **11W**. Alternatively, the convex part **11P** that is formed separately from the sidewall part **11W** may be attached to the inner side of the sidewall part **11W**.

[0135] The cover part **12** is a substantially plate-shaped member corresponding to the bottom part **M1**. As will be described later, the external terminal **20** is attached in advance to the lower surface of the cover part **12** with the gasket **30** (omitted in FIG. **6**) interposed between the external terminal **20** and the cover part **12**.

[0136] First, the positive electrode active material and other materials including, without limitation, the positive electrode binder and the positive electrode conductor are mixed with each other to thereby fabricate a positive electrode mixture. Thereafter, the positive electrode mixture thus fabricated is put into a solvent such as an organic solvent to thereby prepare a positive electrode mixture slurry in paste form. Thereafter, the positive electrode mixture slurry is applied on the two opposed surfaces of the positive electrode current collector **41A** to thereby form the positive electrode active material layers **41B**. Lastly, the positive electrode active material layers **41B** are compression-molded by, for example, a roll pressing machine. In this case, the positive electrode active material layers **41B** may be heated. The positive electrode active material layers **41B** may be compression-molded multiple times. In this manner, the positive electrode **41** is fabricated.

[0137] The negative electrode **42** is fabricated by a procedure similar to the fabrication procedure of the positive electrode **41**. Specifically, the negative electrode current collector **42A** is prepared, following which a negative electrode mixture, which is a mixture of the negative electrode active material and other materials including, without limitation, the negative electrode binder and the

negative electrode conductor, is put into an organic solvent to thereby prepare a negative electrode mixture slurry in paste form. Both end parts, in the width direction, of the negative electrode current collector **42A** are slightly bent in the same direction to thereby form an upper end part **42U** and a lower end part **42L**. Thereafter, the negative electrode mixture slurry is applied on the two opposed surfaces of the negative electrode current collector **42A** to thereby form the negative electrode active material layers **42B**. Thereafter, the negative electrode active material layers **42B** are compression-molded by, for example, a roll pressing machine. In this manner, the negative electrode **42** is fabricated.

[0138] The electrolyte salt is put into the solvent. The electrolyte salt is thereby dispersed or dissolved in the solvent. Thus, the electrolytic solution is prepared.

[0139] First, using a welding method such as a resistance welding method, the positive electrode lead **51** covered at the periphery thereof by the sealant **61** is coupled to the positive electrode **41** (the positive electrode current collector **41A**), and the negative electrode lead **52** is coupled to the negative electrode **42** (the negative electrode current collector **42A**).

[0140] Thereafter, the positive electrode **41** and the negative electrode **42** are stacked on each other with the separator **43** interposed between the positive electrode **41** and the negative electrode **42**, following which the stacked body including the positive electrode **41**, the negative electrode **42**, and the separator **43** is wound to thereby fabricate a wound body **40Z**, as illustrated in FIG. 4. The wound body **40Z** has a configuration similar to the configuration of the battery device **40** except that the positive electrode **41**, the negative electrode **42**, and the separator **43** are each unimpregnated with the electrolytic solution. Note that FIG. 4 omits the illustration of each of the positive electrode lead **51** and the negative electrode lead **52**.

[0141] Thereafter, the wound body **40Z** to which the positive electrode lead **51** and the negative electrode lead **52** are each coupled is placed into the container part **11** through the opening **11K**. In placing the wound body **40Z** in the container part **11**, the wound body **40Z** is placed in a space below the convex part **11P**, that is, a space between the convex part **11P** and the bottom part **11B** in the height direction Z. In this case, the negative electrode lead **52** is coupled to the container part **11** by a welding method such as the resistance welding method. Thereafter, the insulating film **63** is placed on the wound body **40Z**.

[0142] Thereafter, the cover part **12** to which the external terminal **20** is attached in advance with the gasket **30** being interposed between the cover part **12** and the external terminal **20** and on which the insulating film **62** is provided in advance is prepared, following which the positive electrode lead **51** is coupled to the external terminal **20** through the through hole **12K** by a welding method such as the resistance welding method.

[0143] As a result, the wound body **40Z** (the positive electrode **41**) contained inside the container part **11** and the external terminal **20** attached to the cover part **12** are coupled to each other via the positive electrode lead **51**.

[0144] Thereafter, the electrolytic solution is injected into the container part **11** through the opening **11K**. In this case, because the opening **11K** is not closed by the cover part **12** as described above, the electrolytic solution is easily injectable into the container part **11** through the opening **11K** even with the battery device **40** and the external terminal **20** being coupled to each other via the positive electrode lead **51**. The wound body **40Z** including the positive electrode **41**, the negative electrode **42**, and the separator **43** is thereby impregnated with the electrolytic solution. Thus, the battery device **40**, i.e., the electrode wound body, is fabricated.

[0145] Thereafter, the cover part **12** is brought down to approach the container part **11** to thereby close the opening **11K** with the cover part **12**, following which the cover part **12** is welded to the container part **11** by a welding method such as the laser welding method. In this case, as illustrated in FIG. 2, a portion of the positive electrode lead **51** is sandwiched between the cover part **12** and the battery device **40**, and the turning part **513** that is curved is formed on the front side relative to the location where the positive electrode lead **51** is coupled to the external terminal **20**. In this

manner, the outer package can **10** is formed, and the battery device **40** and other components are contained inside the outer package can **10**. Assembly of the secondary battery **1** is thus completed. [0146] The secondary battery **1** after being assembled is charged and discharged. Various conditions including, for example, an environment temperature, the number of times of charging and discharging (the number of cycles), and charging and discharging conditions, may be chosen as desired. As a result, a film is formed on a surface of, for example, the negative electrode **42**. This brings the secondary battery **1** into an electrochemically stable state. The secondary battery **1** is thus completed.

[0147] A description is given next of a method of manufacturing the wiring unit **2** with reference to FIGS. **2** and **3A** to **3C**.

[0148] First, the flexible wiring substrate **21** is fabricated. Specifically, two flexible insulating films are punched into a predetermined shape to thereby obtain the first layer **F1** and the second layer **F2**. Thereafter, the electrically conductive member **32** is attached to a region of the second layer **F2** that is to be the first insulating member **23**, and the electrically conductive member **35** is attached to a region of the second layer **F2** that is to be the second insulating member **26**. Further, the first wiring **W1** and the second wiring **W2** are each formed by, for example, printing on a predetermined region of the second layer **F2**. Thereafter, the first layer **F1** is laid over the second layer **F2** to sandwich the electrically conductive member **32**, the electrically conductive member **35**, the first wiring **W1**, and the second wiring **W2** between the first layer **F1** and the second layer **F2**. Further, the first output terminal **24** to which the first wiring **W1** is to be coupled and the second output terminal **27** to which the second wiring **W2** is to be coupled are formed at an end part of the second layer **F2**.

[0149] Thereafter, the solder layer **33** is applied to the upper surface **32U** and the lower surface **32L** of the electrically conductive member **32**, and the solder layer **36** is applied to the upper surface **35U** and the lower surface **35L** of the electrically conductive member **35**.

[0150] By the above operation, the wiring unit **2** is completed.

[0151] Thereafter, the first coupling terminal **31** of the wiring unit **2** is brought into contact with the surface **20FS** of the external terminal **20** of the secondary battery **1**, following which the first coupling terminal **31** and the external terminal **20** are electrically and mechanically soldered to each other by melting the solder layer **33**. Further, the second coupling terminal **34** of the wiring unit **2** is brought into contact with the lower surface **ILS** of the bottom part **M2** of the outer package can **10** of the secondary battery **1**, following which the second coupling terminal **34** and the lower surface **ILS** of the bottom part **M2** are electrically and mechanically soldered to each other by melting the solder layer **36**.

[0152] By the above operation, the battery unit of an embodiment of the present disclosure is completed.

[0153] As described above, in the battery unit of an embodiment of the present disclosure, the secondary battery **1** is provided with the wiring unit **2**. This makes it possible to electrically couple each of the external terminal **20** serving as the positive electrode terminal and the outer package can **10** serving as the negative electrode terminal in the secondary battery **1** to an external device by the flexible wiring substrate **21** of the wiring unit **2**. Here, the flexible wiring substrate **21** is employed that has a configuration in which the first wiring **W1** and the second wiring **W2** are sandwiched between the first layer **F1** and the second layer **F2**, which are the two flexible insulating films. This makes it possible to achieve a wiring layout having a relatively high degree of freedom while avoiding an electric short circuit between the positive electrode **41** and the negative electrode **42**.

[0154] Further, in the battery unit of an embodiment of the present disclosure, the first coupling terminal **31** is soldered to the external terminal **20** by the solder layer **33**. Here, the solder layer **33** has a structure that is continuous from the upper surface **32U**, through the end surface **32T**, to the lower surface **32L**, of the electrically conductive member **32**, i.e., a structure in which the first part **331**, the second part **332**, and the coupling part **333** are integrated with each other. It is thus

possible, although the wiring unit **2** is small in thickness, to allow the first coupling terminal **31** to be electrically and mechanically coupled to the external terminal **20** firmly. Similarly, the solder layer **36** that solders the second coupling terminal **34** to the bottom part **M2** of the outer package can **10** has a structure that is continuous from the upper surface **35U**, through the end surface **35T**, to the lower surface **35L**, of the electrically conductive member **35**, i.e., a structure in which the first part **361**, the second part **362**, and the coupling part **363** are integrated with each other. It is thus possible, although the wiring unit **2** is small in thickness, to allow the second coupling terminal **34** to be electrically and mechanically coupled to the bottom part **M2** firmly. This makes it possible to avoid coupling failure between the first coupling terminal **31** and the external terminal **20** and coupling failure between the second coupling terminal **34** and the bottom part **M2** due to an external factor such as impact or vibration applied from outside the battery unit. Further, the battery unit of the present embodiment does not include a circuit in, for example, the wiring unit **2**. This makes it unnecessary to consider the influence of heat generated during the soldering on the circuit. [0155] The battery unit of the present embodiment therefore makes it possible to allow for reduction in size and to achieve high reliability.

[0156] Although the present disclosure has been described hereinabove with reference to one or more embodiments including Examples, the configuration of the present disclosure is not limited thereto, and is therefore modifiable in a variety of ways.

[0157] For example, the layout of the flexible wiring substrate of the wiring unit is not limited to the layout of the flexible wiring substrate **21** described in an embodiment described above. FIG. 7 is an exploded perspective diagram illustrating a battery unit **100A** according to a first modification example of an embodiment of the present disclosure. The battery unit **100A** of FIG. 7 includes a wiring unit **2A**. The wiring unit **2A** includes a flexible wiring substrate **21A**. As with the flexible wiring substrate **21** illustrated in FIG. 2, the flexible wiring substrate **21A** includes the first support part **21U**, the second support part **21L**, the middle part **21M**, and the output terminal part **21C**. Note, however, that in the flexible wiring substrate **21A**, the middle part **21M** includes a coupling part **28** and an extraction part **29**. The coupling part **28** couples the first support part **21U** and the second support part **21L** to each other. The extraction part **29** branches from the coupling part **28**. The output terminal part **21C** is provided on an end part of the extraction part **29** opposite to the coupling part **28**. The coupling part **28** and the extraction part **29** are integrated with each other and includes the stacked structure of the first layer **F1** and the second layer **F2** similarly to the first insulating member **23**. The coupling part **28** and the extraction part **29** are provided with the first wiring **W1** and the second wiring **W2**, which are sandwiched between the first layer **F1** and the second layer **F2**. In the example of the wiring unit **2A** illustrated in FIG. 7, the first wiring **W1** extends from the first coupling terminal **31**, through the coupling part **28** and the extraction part **29**, to the first output terminal **24**. The second wiring **W2** extends from the second coupling terminal **34**, through the coupling part **28** and the extraction part **29**, to the second output terminal **27**.

[0158] Further, in the battery unit **100** of an embodiment described above, the first coupling terminal **31** of the wiring unit **2** is joined to the upper surface **1US** of the secondary battery **1**, and the second coupling terminal **34** of the wiring unit **2** is joined to the lower surface **ILS** of the secondary battery **1**; however, the present disclosure is not limited thereto. For example, as with a battery unit **100B** according to a second modification example of an embodiment of the present disclosure illustrated in FIGS. 8A and 8B, the first coupling terminal **31** of a wiring unit **2B** may be joined to the upper surface **1US** of the secondary battery **1**, and the second coupling terminal **34** of the wiring unit **2B** may be joined to the side surface **1SS** of the secondary battery **1**. In such a case also, the battery unit **100B** achieves effects similar to those of the battery unit **100** of an embodiment described above. Note that FIG. 8A is a perspective diagram illustrating an overall configuration example of the battery unit **100B** according to the second modification example of an embodiment of the present disclosure. FIG. 8B is an exploded perspective diagram illustrating an overall configuration example of the battery unit **100B**. The battery unit **100B** includes the wiring

unit **2B**. The wiring unit **2B** includes a flexible wiring substrate **21B** including, for example, a stem part **25** having a generally L-shape. The second coupling terminal **34** is provided at a first end part of the stem part **25**, and the output terminal part **21C** is provided at a second end part of the stem part **25**. The first insulating member **23** is coupled to a region, of the stem part **25**, between the second coupling terminal **34** and the output terminal part **21C** via the coupling part **28**. The first insulating member **23** supports the first coupling terminal **31**. The wiring unit **2B** is configured such that a boundary part between the first insulating member **23** and the coupling part **28** and a boundary part between the coupling part **28** and the stem part **25** are each bent substantially perpendicularly, thereby allowing the first coupling terminal **31** to face the upper surface **1US** of the secondary battery **1** and the second coupling terminal **34** to face the side surface **1SS** of the secondary battery **1**.

[0159] Further, in the battery unit of an embodiment described above, the description has been given of the case where the outer package can **10** is a welded can (a crimpless can); however, the outer package can is not particularly limited in configuration, and may be a crimped can which has undergone crimping processing. In the crimped can, a container part and a cover part separate from each other are crimped to each other with a gasket interposed between the container part and the cover part.

[0160] Further, although the description has been given of the case where the electrode reactant is lithium, the electrode reactant is not particularly limited. Accordingly, the electrode reactant may be another alkali metal such as sodium or potassium, or may be an alkaline earth metal such as beryllium, magnesium, or calcium, as described above. In addition, the electrode reactant may be another light metal such as aluminum.

[0161] Further, in the battery unit of an embodiment described above, the description has been given of the case where the battery unit includes the secondary battery as the battery; however, the present disclosure is not limited thereto, and the battery unit may include a primary battery. FIG. **9** is a perspective diagram illustrating a configuration example of a battery unit **100C** according to a modification example of the present disclosure, and the battery unit **100C** includes a primary battery **9** instead of the secondary battery **1**. The battery unit of an embodiment described above includes the secondary battery **1** and the wiring unit **2**. The wiring unit **2** includes the flexible wiring substrate **21**. The flexible wiring substrate **21** is provided with the first coupling terminal **31** and the second coupling terminal **34**. In contrast, the battery unit **100C** of FIG. **9** includes a wiring unit **90** instead of the wiring unit **2**.

[0162] FIG. **10** is an exploded perspective diagram illustrating a configuration example of the wiring unit **90** illustrated in FIG. **9** in an unbent state. The wiring unit **90** includes, for example, a flexible wiring substrate **91**. The flexible wiring substrate **91** is provided with a first coupling terminal **81** and a second coupling terminal **84**. The first coupling terminal **81** is an electrically conductive member that is disposed to face a positive electrode terminal **9A** of the primary battery **9** and is electrically joined to the positive electrode terminal **9A** by, for example, soldering. The second coupling terminal **84** is an electrically conductive member that is disposed to face a bottom surface **9B** of the outer package can and is electrically joined to the bottom surface **9B** of the outer package can by, for example, soldering. The bottom surface **9B** of the outer package can serves as a negative electrode terminal of the primary battery **9**. The flexible wiring substrate **91** includes a first support part **91U**, a second support part **91L**, a middle part **91M**, and an output terminal part **91C**.

[0163] The first support part **91U** supports the first coupling terminal **81**. The first support part **91U** is provided to be opposed to the positive electrode terminal **9A** exposed on a surface of the primary battery **9**. The second support part **91L** supports the second coupling terminal **84**. The second support part **91L** is provided to be opposed to the bottom surface **9B** of the primary battery **9**. The middle part **91M** couples the first support part **91U** and the second support part **91L** to each other. The middle part **91M** is provided along a side surface of the primary battery **9**. Note that, although

FIG. 10 illustrates the flexible wiring substrate **91** in an unbent state where all the first support part **91U**, the second support part **91L**, and the middle part **91M** extend along the same plane, in the battery unit **100C**, a boundary part between the first support part **91U** and the middle part **91M** is bent substantially perpendicularly, and a boundary part between the second support part **91L** and the middle part **91M** is bent substantially perpendicularly. The primary battery **9** has a height **H1** lower than the height **H** of the secondary battery **1**. Accordingly, in the battery unit **100C** including the primary battery **9**, a length of the middle part **91M** of the flexible wiring substrate **91** is shorter than a length of the middle part **21M** of the flexible wiring substrate **21** (FIG. 2).

[0164] As illustrated in FIG. 10, the flexible wiring substrate **91** further includes the first wiring **W1** and the second wiring **W2**. The first wiring **W1** extends from the first coupling terminal **81**, sequentially through the middle part **91M** and the second support part **91L**, to a first output terminal **97**. The second wiring **W2** extends from the second coupling terminal **84** to a second output terminal **94**. The first wiring **W1** and the second wiring **W2** are each a printed wiring including highly electrically conductive material such as copper. The first wiring **W1** and the second wiring **W2** are sandwiched between, for example, two flexible insulating films (that are each a polyimide film, for example).

[0165] The first support part **91U** includes a first insulating member **93** that holds the first coupling terminal **81**. The middle part **91M** couples the first support part **91U** and the second support part **91L** to each other. The middle part **91M** is integrated with the first support part **91U** and the second support part **91L**. The middle part **91M** is provided along a side surface of the secondary battery **1**. The second support part **91L** includes a second insulating member **96** that holds the second coupling terminal **84**.

[0166] In the battery unit **100C** illustrated in FIGS. 9 and 10, an electromotive force of the primary battery **9** may be extracted from the output terminal part **91C**.

[0167] The effects described herein are mere examples, and effects of the technology are therefore not limited to those described herein. Accordingly, the technology may achieve any other effect.

[0168] The present disclosure may encompass the following embodiments.

1 [0169] A battery unit including: [0170] a battery including a first electrode terminal and a second electrode terminal; and [0171] a wiring unit including [0172] a flexible wiring substrate, [0173] a first coupling terminal that is provided on the flexible wiring substrate and is to be joined to the first electrode terminal, and [0174] a second coupling terminal that is provided on the flexible wiring substrate and is to be joined to the second electrode terminal, in which [0175] at least one of the first coupling terminal or the second coupling terminal includes an electrically conductive member and a solder layer, [0176] the electrically conductive member has a first surface, a second surface, and a through hole, the through hole extending from the first surface to the second surface, and [0177] the solder layer includes a first part, a second part, and a coupling part, the first part covering all or a part of the first surface of the electrically conductive member, the second part covering all or a part of the second surface of the electrically conductive member, the coupling part passing through the through hole and coupling the first part and the second part to each other.

2 [0178] The battery unit according to <1>, in which the battery includes [0179] an external terminal serving as the first electrode terminal, and [0180] an outer package can serving as the second electrode terminal, the outer package can including a bottom part and a sidewall part, the bottom part being opposed to the external terminal, the sidewall part being provided upright along an outer edge of the bottom part.

3 [0181] The battery unit according to <2>, further including [0182] an insulating layer provided between the external terminal and the wiring unit, in which [0183] the external terminal is surrounded by an upper end part of the sidewall part, [0184] the insulating layer has an opening, and [0185] the external terminal and the first coupling terminal are soldered to each other through the opening.

4 [0186] The battery unit according to any one of <1> to <3>, in which the flexible wiring substrate

includes [0187] a first support part supporting the first coupling terminal, [0188] a second support part supporting the second coupling terminal, [0189] a middle part coupling the first support part and the second support part to each other, [0190] a first wiring extending from the first coupling terminal, through the middle part, to the middle part, and [0191] a second wiring extending from the second coupling terminal, through the middle part, to the middle part.

5 [0192] The battery unit according to any one of <1>to <4>, in which [0193] the battery includes [0194] a cover part that holds the first electrode terminal, and [0195] a container member serving as the second electrode terminal, the container member including a bottom part and a sidewall part, the bottom part being opposed to the cover part, the sidewall part being provided upright along an outer edge of the bottom part, and the middle part is provided along the sidewall part of the battery.

6 [0196] A battery unit that is a wiring unit for a battery, the wiring unit being mountable on the battery including a first electrode terminal and a second electrode terminal, the battery unit including: [0197] a flexible wiring substrate; [0198] a first coupling terminal that is provided on the flexible wiring substrate and is to be joined to the first electrode terminal; and [0199] a second coupling terminal that is provided on the flexible wiring substrate and is to be joined to the second electrode terminal, in which [0200] at least one of the first coupling terminal or the second coupling terminal includes an electrically conductive member and a solder layer, [0201] the electrically conductive member has a first surface, a second surface, and a through hole, the through hole extending from the first surface to the second surface, and [0202] the solder layer includes a first part, a second part, and a coupling part, the first part covering all or a part of the first surface of the electrically conductive member, the second part covering all or a part of the second surface of the electrically conductive member, the coupling part passing through the through hole and coupling the first part and the second part to each other.

[0203] It should be understood that various changes and modifications to the embodiments described herein will be apparent to those skilled in the art. Such changes and modifications can be made without departing from the spirit and scope of the present subject matter and without diminishing its intended advantages. It is therefore intended that such changes and modifications be covered by the appended claims.

Claims

1. A battery unit comprising: a battery including a first electrode terminal and a second electrode terminal; and a wiring unit including a flexible wiring substrate, a first coupling terminal that is provided on the flexible wiring substrate and is to be joined to the first electrode terminal, and a second coupling terminal that is provided on the flexible wiring substrate and is to be joined to the second electrode terminal, wherein at least one of the first coupling terminal or the second coupling terminal includes an electrically conductive member and a solder layer, the electrically conductive member has a first surface, a second surface, and a through hole, the through hole extending from the first surface to the second surface, and the solder layer includes a first part, a second part, and a coupling part, the first part covering all or a part of the first surface of the electrically conductive member, the second part covering all or a part of the second surface of the electrically conductive member, the coupling part passing through the through hole and coupling the first part and the second part to each other.

2. The battery unit according to claim 1, wherein the battery includes an external terminal serving as the first electrode terminal, and an outer package can serving as the second electrode terminal, the outer package can including a bottom part and a sidewall part, the bottom part being opposed to the external terminal, the sidewall part being provided upright along an outer edge of the bottom part.

3. The battery unit according to claim 2, further comprising an insulating layer provided between the external terminal and the wiring unit, wherein the external terminal is surrounded by an upper

- end part of the sidewall part, the insulating layer has an opening, and the external terminal and the first coupling terminal are soldered to each other through the opening.
- 4.** The battery unit according to claim 1, wherein the flexible wiring substrate includes a first support part supporting the first coupling terminal, a second support part supporting the second coupling terminal, a middle part coupling the first support part and the second support part to each other, a first wiring extending from the first coupling terminal, through the middle part, to the middle part, and a second wiring extending from the second coupling terminal, through the middle part, to the middle part.
- 5.** The battery unit according to claim 4, wherein the battery includes a cover part that holds the first electrode terminal, and a container member serving as the second electrode terminal, the container member including a bottom part and a sidewall part, the bottom part being opposed to the cover part, the sidewall part being provided upright along an outer edge of the bottom part, and the middle part is provided along the sidewall part of the battery.
- 6.** A wiring unit for a battery, the wiring unit being mountable on the battery including a first electrode terminal and a second electrode terminal, the wiring unit comprising: a flexible wiring substrate; a first coupling terminal that is provided on the flexible wiring substrate and is to be joined to the first electrode terminal; and a second coupling terminal that is provided on the flexible wiring substrate and is to be joined to the second electrode terminal, wherein at least one of the first coupling terminal or the second coupling terminal includes an electrically conductive member and a solder layer, the electrically conductive member has a first surface, a second surface, and a through hole, the through hole extending from the first surface to the second surface, and the solder layer includes a first part, a second part, and a coupling part, the first part covering all or a part of the first surface of the electrically conductive member, the second part covering all or a part of the second surface of the electrically conductive member, the coupling part passing through the through hole and coupling the first part and the second part to each other.
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